Fitting absolutely monotonic real functions in bounded intervals

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We discuss the problem of fitting absolutely monotonic functions to noisy random samples in bounded intervals. This specific shape-constrained estimation problem has not been significantly explored in the literature, though it is related to other kinds of shape-constrained estimation such as fitting exponential sums or convex polynomial fitting. A wide variety of potential approaches applicable to the problem are discussed, including various forms of exponential sum fitting, using Bernstein polynomial bases or splines in constrained optimization problems, and non-linear least squares optimization with Vandermonde matrices. A novel approach, Greedy Lowballing Absolutely Monotonic Polynomial Fitting (GLAMPF) is also introduced. This method finds monomial terms to iteratively approximate the sampled data from below (lowballing), ensuring absolute monotonicity by construction and robustness to a variety of function shapes. However, it lacks general optimality and occasionally underestimates certain regions of the sample. A few of the most promising methods were implemented and compared on generated samples from a wide variety of absolutely monotonic and convex functions. While convex splines displayed the best results in cases where low degree polynomials were sufficient, GLAMPF was the most robust approach with the strongest guarantees and produced acceptable results in every test case.

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1 Introduction

1.1 Motivating problem: reward curves

The motivating problem for this paper is the production of *reward curves from scores in a competitive single player game*. Specifically, Beat Saber[1] is a virtual reality rhythm game released in 2018 in which players play single player maps with a maximum possible score, attempting to get as close as possible. Scores in Beat Saber are often represented as percentages of the maximum possible score. Beat Saber has rich scoring mechanics[2], and a healthy competitive scene with upwards of 60,000 active ranked players[3], which can be observed through its two most popular ranked leaderboards:

[1] `https://beatsaber.com/`

[2] `https://bsmg.wiki/ranking-guide.html`

[3] Verified using BeatLeader's search functionality at `https://beatleader.com/ranking/1?mapsType=all&recentScoreTime=1721433600`

ScoreSaber[4] and BeatLeader[5]. Competitive Beat Saber focuses on large pools of *custom* maps made by members of the community (*mappers*) that undergo a ranking process[6] to ensure their competitive viability. For example, BeatLeader offers over 3500 different ranked maps[7].

Experienced Beat Saber players observe a high diminishing returns difficulty curve. The closer to 100%, the harder it is to improve the score further. There are exactly two 100% scores across all ranked maps in Beat Leader (over 3500 maps, over 60,000 players), on particularly easy maps by particularly skilled players. The difference in difficulty between a 98% and a 99% score in a map is orders of magnitude larger than the difference between a 90% and a 91%. Most scores that do not fail the map are over 70% to begin with.

Different maps do not just have different difficulties, but also different difficulty *curves*, with the difficulty accumulating at different relative ranges of the score spectrum, while always preserving the diminishing returns nature. The problem of finding adequate reward curves for each map is difficult and highly sought-after by the Beat Saber ranked community [Casanova, 2025]. Shape-constrained fitting from player scores is a valuable tool in this context. Naturally, a higher score should always give a higher reward (often quantified as Performance Points (PP)), introducing a growing constraint on the reward curve. Moreover, the diminishing returns nature justifies introducing further growth constraints that increase the rate at which the rewards increase. While convexity constraints could be used, there are growing convex curves that do not adequately reflect the ever increasingly difficult nature of approaching 100% score in Beat Saber maps (see, for example, figure 1). Therefore, we instead consider the more constraining problem of fitting smooth absolutely monotonic curves from sample scores.

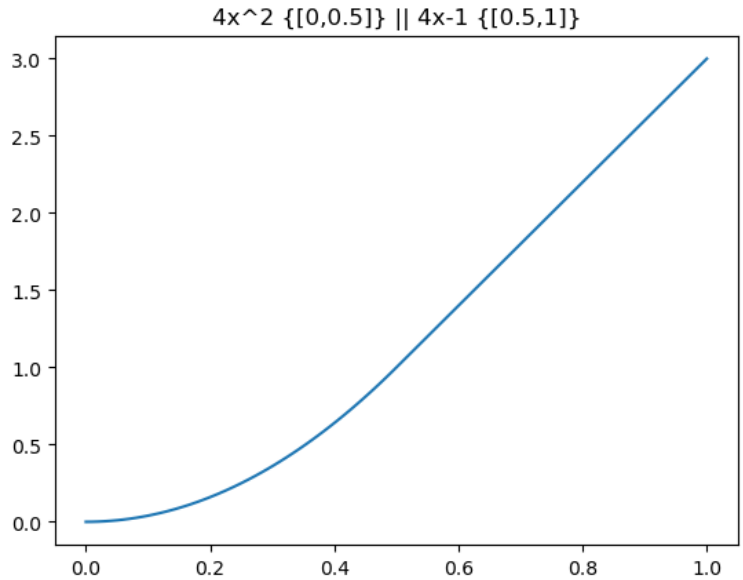
1.2 Absolutely monotonic functions

A real function is *absolutely monotonic* if all of its derivatives are nonnegative. Absolutely monotonic functions grow faster and faster as we approach higher values, curving upwards in every sense. They are growing and convex, though not all growing convex functions are absolutely monotonic.

Convex smooth functions, and in particular fitting convex smooth functions from samples, have been studied extensively [Groeneboom et al., 2001,

[4] `https://scoresaber.com/`
[5] `https://beatleader.xyz/`
[6] `https://beatleader.wiki/en/ranking/Ranking-your-map`
[7] `https://beatleader.xyz/leaderboards`

Figure 1: Smooth piecewise polynomial that is growing and convex but not absolutely monotonic



Lu, 2015, Lebair and Shen, 2015, Nagahara and Martin, 2013, Zhang, 2004, He and Shi, 1998, Aguilera and Morin, 2008, Hannah and Dunson, 2013, Holmström and Peterss Lachand-Robert and Oudet, 2005, Kopotun et al., 2011, Wang and Ghosh, 2012, Ghosal et al., 2023] due to the multitude of applications in economics, statistics, machine learning, etc. Like other types of *shape-constrained estimation*, we know or wish the resulting function to fulfill certain growth or smoothness properties, but also wish to fit the available data as best as possible.

Absolutely monotonic shape-constrained estimation from fixed samples has not been studied as extensively. In general, the majority of methods that ensure absolute monotonicity of the resulting functions do so *by construction*, and usually as a side effect of tighter restrictions. For example, fitting exponential sums [Holmström and Petersson, 2002, Keller and Plonka, 2021, Chunaev and Safiullin, 2020, Evans et al., 1980, Smith et al., 1976, Kammler, 1976]. These also sometimes presume we can select the points to sample on.

In §2 we discuss many potential approaches to the motivating problem in detail, highlighting this tension between less constrained methods that do not guarantee absolute monotonicity, and methods that do guarantee absolute monotonicity but are not very well suited for fitting from random samples.

1.3 Problem statement

Given a single variable **real valued** set of datapoints in a **bounded interval** (we can assume $[0, 1]$ without loss of generality), we consider the problem of finding **absolutely monotonic** functions that best approximate the data. We will, however, compare with some methods that only guarantee convexity due to their prominence and to understand their relative advantages and disadvantages.

We presume noise in the data. In the Beat Saber map context, the x axis indicates the score obtained in the map by the player, and the y axis indicates the reward associated with it, which in the data is represented by the player skill of the player that produced the score. The player skill of a player in a ranked setting represents *their best possible performance*, meaning that better players can sometimes produce worse scores (e.g. not taking the map seriously, being unlucky, or having less than ideal circumstances), but it is far more unlikely that a player will produce a score above their skill level. Moreover, players can attempt maps many times, retaining only their best scores. Therefore, we presume noise in the data will have a strong positive bias (the skill of the player is higher than this score in this map would suggest). In other words, our estimation should ideally favour functions that **underestimate** the data while being as accurate as possible. In order to achieve this, we use asymmetric loss functions in our methods. We consider two different asymmetric loss functions:

1. Pinball loss

$$l_1(r) = \begin{array}{ll} \tau r & \text{if } r \geq 0 \\ (\tau - 1)r & \text{if } r < 0 \end{array} \tag{1}$$

2. Soft floor

$$l_2(r) = |r| + e^{-\tau r} \tag{2}$$

One final aspect to consider is how we will use the resulting estimations. Our goal is to use real score data to learn how to best describe the reward curves of different maps. The results of this could then be used to train a different model that looks at the map gameplay properties themselves to estimate those parameters without needing scores. This is useful for newly ranked maps that do not have enough scores yet but need a reward curve. Therefore, we are aiming for a **(semi-)parametric** method that can output a bounded and small number of parameters that represents the fitted curve.

We can summarize these properties in a single problem statement. The problem statement leaves some room for variability to consider the different approaches' advantages and disadvantages.

Problem 1.1. *Given a set of real valued datapoints* $\{x_i, y_i\}$*, where* $x_i \in [0, 1]$*, find a function* $f \in \mathcal{C}^\infty([0, 1])$ *that:*

- *is absolutely monotonic in* $[0, 1]$*.*
- *approximates the datapoints as best as possible with a preference for underestimation.*
- *can be represented (semi-)parametrically with a bounded, low number of parameters.*

The remainder of this document discusses several potential approaches to solving this problem (or slightly relaxed versions of it), discussing their advantages and disadvantages. We also introduce a novel approach, Greedy Lowballing Absolutely Monotonic Polynomial Fitting (GLAMPF), specifically tailored for this problem. We implemented and compared some of these approaches with simulation studies.

§2 explains the approaches, with their pros and cons, including the implementation of those that were implemented. §3 presents the simulation and evaluation set up for the comparison of the implemented approaches. §4 discusses the results of the evaluation and general discussion on how different methods compare. §5 presents the conclusions from this paper.

2 Potential approaches

We discuss a number of reasonable approaches to the presented problem, with their rationales, advantages, and disadvantages. A summary table can be found at the end of this section.

2.1 Positive sums of positive exponentials

Exponential functions with positive exponents are always absolutely monotonic. A deeply related family of functions are *completely monotonic functions*, which are functions with alternating sign derivatives. It is easy to see that $f(x)$ is absolutely monotonic in (a, b) if and only if $f(-x)$ is completely monotonic in $(-a, -b)$. Moreover, Bernstein's theorem characterizes all completely monotonic functions in $[0, +\infty)$, and consequently also all absolutely monotonic functions in $(-\infty, 0]$, as sums of exponentials:

Theorem 2.1 ([Bernstein, 1929])**.** *Let* $f \in \mathcal{C}^\infty([0, +\infty))$*. Then,* f *is completely monotonic if and only if it is the Laplace transform of a non-negative*

finite Borel measure μ. Explicitly:

$$f(x) = \int_0^\infty e^{-tx} d\mu(t) \tag{3}$$

or, rephrasing it for absolutely monotonic functions:

Theorem 2.2 ([Bernstein, 1929])**.** *Let $f \in \mathcal{C}^\infty((-\infty, 0])$. Then, f is absolutely monotonic if and only if there is a non-negative finite Borel measure μ such that:*

$$f(x) = \int_0^\infty e^{tx} d\mu(t) \tag{4}$$

In a few words, any absolutely monotonic function in $(-\infty, 0]$ can be uniquely characterised as an infinite positive sum of exponentials with positive exponents. This also forces f to be analytic because the representation is analytic. The mathematical theory of completely monotonic, absolutely monotonic, and related families of functions is much broader and deeper than what we discuss or use in this paper[Koumandos, 2014].

This is obviously interesting from the point of view of attempting semi-parametric fitting of absolutely monotonic functions: if a function can be expressed as a sum of exponentials, can we select the most significant exponents in this sum to estimate the function? There is one important initial problem with this approach: it only applies to functions that are absolutely monotonic in all of $(-\infty, 0]$. Many absolutely monotonic functions in $[0, 1]$ cannot be extended analytically to $(-\infty, 0]$, and therefore do not admit an exponential sum representation. A simple and important example of this is monomials $f(x) = x^n$. All of these are absolutely monotonic (and, in fact, analytic) in $[0, 1]$, but cannot be extended to an absolutely monotonic function in $(-\infty, 0]$, and cannot be represented as sums of exponentials.

When attempting to use sums of exponentials to fit datapoints to absolutely monotonic functions, this severely limits the representation power. Nonetheless, should we accept this limitation, there is extensive literature on the subject of fitting exponential sums.

Prony's method relies on sampling at specific intervals to exploit the structure of an exponential sum[Holmström and Petersson, 2002, Keller and Plonka, 2021, Chunaev and Safiullin, 2020]. This is an effective method for calculating transforms of known functions that can be sampled at prescribed intervals, but is not suitable for fitting a function to random data that does not adjust to this structure and contains noise.

Nonlinear least squares optimization with various suitable setups can be used to approximate a set of parameters (both coefficients and exponents) for an exponential sum that minimizes loss on the datapoints[Holmström and Petersson, 2002, Evans et al., 1980]. This presents challenges such as convergence to local optima (particularly for unusual loss functions), high reliance on the parameterization choice, complex and unpredictable effects of noise in the data, the requirement for the loss function to be smooth, and poor computational scalability to higher number of parameters. It is nonetheless a feasible option for datapoints that can be approximated using exponential sums.

A number of other approaches and variations on the aforementioned approaches for fitting exponential sums exist. All of these share some of the challenges discussed, especially the need for specific sampling, sensitivity to noise, convergence to local optima, and computational issues. For example, [Smith et al., 1976] uses a spectral method to fit exponential sums, but the approach is outdated, computationally problematic, shows mixed results, and most importantly, requires admitting negative coefficients, losing absolute monotonicity. It is possible that the novel greedy lowballing polynomial method introduced in §2.3 can be used with exponential sums as well, but it would incur the aforementioned representation limitation and thus this was not explored.

2.2 Constrained piecewise linear interpolation

While exponential sums ensure absolute monotonicity *by construction*, an opposite approach uses *explicit constraints* to enforce monotonicity. An extreme version of this is to build piecewise linear interpolations that minimize loss with respect to the data points, under, for example, convexity constraints. Convexity can be expressed as optimization constraints by indicating that the second order discrete differences must all be positive:

$$\frac{y_{i+2} - y_{i+1}}{x_{i+2} - x_{i+1}} - \frac{y_{i+1} - y_i}{x_{i+1} - x_i} \geq 0 \tag{5}$$

Piecewise linear interpolation has high flexibility, and therefore can produce low loss fits, and is achievable efficiently with a wide variety of optimization algorithms, but it has a number of important issues. First, higher-order monotonicity constraints involve more and more complex optimization constraints, with strict absolute monotonicity not being possible to express (but rather up to an arbitrary degree of discrete differences). The more higher-order constraints are used, the less the computational attractiveness of the

optimization algorithm can be exploited. Moreover, piecewise linear interpolation is only continuous and not differentiable. Finally, this method is inherently nonparametric, with the result being dependent on the number of interpolating points used, meaning that using a small number of parameters will lead to worse fits.

We implemented this method for comparison purposes even if it does not satisfy important conditions of our problem.

2.3 Polynomials

Taylor's theorem is foundational in real analysis and naturally connected to the problem of fitting absolutely monotonic functions. If we restrict our potential functions to analytic functions $\mathcal{C}^\omega$, then the Taylor series at any point of the function explicitly presents all the derivatives of the function at that point. Moreover, if all derivatives at a point are positive and the function is analytic, it is guaranteed that they will only grow, and therefore the function is absolutely monotonic in the whole half-line after that point. We can summarize this in the following lemma, by building analytic functions from their Taylor series at 0:

Lemma 2.1. *An analytic function $f \in \mathcal{C}^\omega([0, +\infty))$ is absolutely monotonic in the positive half-line if and only if it is of the form $\sum_{i=0}^{\infty} a_i x^i$, where every $a_i \geq 0$.*

An important result justifies the restriction to analytic functions to perform approximations on a compact interval $[0, 1]$. An absolutely monotonic function in $[0, 1]$ in particular is absolutely monotonic in $(0, 1)$, and therefore it must be analytic in that open interval [Koumandos, 2014]. While analyticity may not extend to the endpoints, this is enough to use the Weierstrass theorem to ensure that the function can be uniformly approximated by absolutely monotonic polynomials. We can summarize all of this in the following lemma:

Lemma 2.2. *A function $f \in \mathcal{C}^\infty([0, 1])$ is absolutely monotonic if and only if it can be uniformly approximated by polynomials of the form $\sum_{i=0}^{N} a_i x^i$, where every $a_i \geq 0$.*

In short, we can use non-negative polynomial sums to fit our data, ensuring absolute monotonicity by construction and strong guarantees about

approximation capability.

Some important questions remain, and different polynomial fitting methods will be discussed in this section to address these questions. The first is what method to use to obtain good approximation coefficients a_i from the sample without further knowledge about the underlying function (other than absolute monotonicity), especially when the sample is predetermined and the function cannot be sampled at specific points. This also relates to the question of how the fitting method responds to noisy data. Finally, there is the question of how many coefficients are necessary to provide a good approximation of the function and how to provide as good an approximation as possible with as few as possible (or a predefined number of) coefficients.

One thing worth discarding from the beginning is spectral methods. Taylor series are sensitive to small variations and overall lack a stable structure that preserves coefficients and/or degree dominance under approximation circumstances, an issue that grows the further away from 0 that we evaluate the series. From a practical point of view, it is not feasible, especially from sampled noisy data, to reliably reconstruct an exact and/or uniquely defined Taylor series, or to isolate the largest coefficients in the Taylor series. Taylor polynomials with very different coefficient distributions can be very similar pointwise, meaning that there is no way to define a preferred coefficient or degree from data points alone.

2.3.1 Bernstein polynomials

In certain ways the closest approach to spectral methods, Bernstein polynomials offer a promising approach to our problem. Work like [Wang and Ghosh, 2012, Ghosal et al., 2023] present explicit methods to perform shape-constrained regression using Bernstein polynomials from sampled data, with strong approximation and performance guarantees. Specifically, Bernstein basis polynomials are a basis for the vector space of polynomials of a certain degree with important interpolation properties. [Wang and Ghosh, 2012] showed that polynomial regression of a certain degree with shape constraints can be solved by using an adequate matrix representing the shape constraints (monotonicity, convexity, etc.) on the discrete differences applied to the Bernstein basis. With a typical squared residual objective this can then be solved using linear least squares optimization.

However, this has a number of challenges for application to our problem. First, no matrix expression for absolute monotonicity can be found in the

literature. Nonetheless, we believe it would be possible to express absolute monotonicity up to the maximum derivative that matters with adequate matrices, but these would likely increase the size of the optimization problem substantially, as it involves explicit constraints rather than absolute monotonicity by construction. More importantly, the method is constrained from inception to a maximum degree of polynomials. In absolutely monotonic fitting, especially when the underlying function is (hyper-)exponential, we may need very high degree polynomials to adequately represent the growth pattern, even for small samples. The Bernstein polynomial linear method requires a pre-determined maximum degree, and the complexity of the optimization problem increases with the degree (so large degrees are not feasible). Finally, this is inherently a nonparametric method that requires a quite large set of coefficients dependent on the degree. The basis is not set up in a way that allows removing some of the elements while preserving a significant proportion of fitting capabilities.

Another approach using Bernstein polynomials would be feasible if the function could be sampled at specific intervals, following Bernstein's original proof of the Weierstrass approximation theorem [Bernstein, 1912], but as discussed this is also infeasible for our problem.

2.3.2 Constrained least squares Vandermonde

A simple yet important to compare to approach uses the Vandermonde matrix over the dataset up to a maximum degree to present a linear optimization problem to find a polynomial estimation. However, in order to enforce absolute monotonicity, non-negativity constraints need to be enforced on the matrix. This makes linear least squares unsuitable to solve the problem due to the severe restriction of the parameter space. Furthermore, the use of a custom loss function in our problem forces the use of nonlinear least squares optimization. Moreover, high degree polynomials are required to represent absolutely monotonic functions well enough, increasing the complexity of the optimization problem. Perhaps much more importantly, Vandermonde interpolation is numerically unstable and badly represented for high degrees, with high sensitivity to noise or numeric imprecisions during optimization.

We implemented this method for comparison purposes even if it does not satisfy important conditions of our problem.

2.3.3 Convex splines

A popular family of shape-constrained methods discussed extensively in the literature use spline interpolation [Lu, 2015, Lebair and Shen, 2015, Nagahara and Martin, 2013, Zhang, 2004, He and Shi, 1998]. Similar to Bernstein polynomials, these methods use explicit constraints on an optimization problem to ensure monotonicity constraints. They have similar issues to Bernstein polynomials for application to our problem. Absolute monotonicity constraints are not discussed in the literature and would be even harder to implement for spline interpolation, if possible at all. Degree constraints are even harsher than Bernstein polynomial methods, with a degree of 3 used frequently in the literature, being entirely unsuitable for absolutely monotonic functions like (hyper-)exponentials. They are also nonparametric with the number of parameters being highly correlated with the quality of the approximation.

This method is more promising in terms of approximation capability and performance than other methods and relatively easy to implement, so we implemented it with degree 3 and convexity constraints for comparison, even if it does not satisfy all of the conditions of our problem.

2.3.4 Greedy lowballing monomial sums

To the best of our knowledge, there is no significant literature on the use of greedy algorithms for polynomial fitting. To a large degree, this makes sense: the general polynomial fitting problem is unstable and must be solved holistically for satisfactory solutions, lacking any significant stable structure that can be exploited to iteratively build approximations that consistently improve. Moreover, without further constraints, global solvers, including some of the ones discussed so far, work efficiently and effectively, so there is no reason for slower methods with fewer guarantees.

However, in the particular context of the problem presented in this paper, we argue that a greedy approach is attractive. From the point of view of the (semi-)parametric goal, producing a polynomial one monomial at a time slowly approaching the target allows us to select a limited number of monomials to include in the final solution, reducing the quality of the approximation in a measured way in order to obtain a simpler solution expression. Perhaps more importantly, the absolutely monotonic constraints, together with the asymmetric loss favouring underestimation, lend themselves to an iterative *lowballing* approach, where we slowly add terms, always underestimating the solution by smaller and smaller amounts at each step.

We thus introduce the Greedy Lowballing Absolutely Monotonic Polynomial Fitting (GLAMPF) algorithm. This algorithm is highly specialised to this problem and has significant disadvantages, especially if used for more general tasks, compared to the other methods discussed in this section, but is particularly attractive to our problem in particular. We can summarize its key features as follows:

- The algorithm has two sets of parameters: *degrees* and *coefficients*, which are optimized simultaneously. Their number is not necessarily predetermined but can be limited with measured effects, making the method semi-parametric.
- At each iteration, we add a new monomial with a certain degree and coefficient, of the form $c_j x^{d_j}$, that minimizes the loss with respect to the remainder of the target. The same degree can be added multiple times. After each iteration, we subtract the current monomial from the target value at each point.
- Once an optimal degree and coefficient has been found for a given iteration, we reduce its contribution by a factor $\gamma < 1$ before subtracting it, to ensure the *lowballing*. Together with the asymmetric loss, this ensures that the target is always underestimated, but built towards in steps that best represent the structure of the remainder. That is, we do ${}^{j+1}y_i = {}^{j}y_i - \gamma c_j x_i^{d_j}$.
- γ is increased at each iteration, approaching 1 over time, to slowly converge to a full solution.
- We accumulate the coefficients and degrees, and once a certain threshold of approximation or a threshold for small coefficients are surpassed, the algorithm stops. We filter excessively small resulting coefficients and/or limit to a certain number of highest coefficients for an efficient parametric final expression. We note that $0^d = 0$ and $1^d = 1$ for all d, which means that the coefficients of different degrees are comparable in scale when fitting in the $[0, 1]$ interval.

Among important disadvantages of this approach compared to other discussed ones, we include computational inefficiency due to the greedy nested optimization pattern and dependence on hyperparameter choice and data particularities in the overall result. However, unlike all the other discussed methods, this method simultaneously ensures absolute monotonicity by construction, behaves particularly well with asymmetric loss due to its inherent

asymmetric nature, and produces semi-parametric outputs that can easily be limited to a small number of parameters with measured effects.

We implemented this method and consider it the one to best satisfy our particular problem, having potential for use in other variations, while having significant disadvantages outside of its scope.

2.4 Summary

Method	**Representation**	**Monotonicity**	**Optimality**	**Hyperparams**	**Noise**	**Loss**	**Sampling**	**Parametric**	**Performance**	**Complexity**
Prony methods §2.1	⚠ Exponential only	✅ Absolutely monotonic, by construction	✅ Yes	✅ Irrelevant	❌ Highly sensitive	❌ Fixed target	❌ Fixed	✅ Parametric	✅ Fast	🔵 Medium
Non-linear least squares exponential sums §2.1	⚠ Exponential only	✅ Absolutely monotonic, by construction	⚠ Local optima	⚠ Sensitive	⚠ Sensitive	🔵 Flexible, but affected	✅ Any	✅ Parametric	🔵 Moderate	✅ Low
Constrained piecewise linear interpolation§2.2	❌ Not differentiable	⚠ Convex	✅ Yes	✅ Irrelevant	🔵 Robust	✅ Flexible	✅ Any	❌ Nonparametric	✅ Fast	✅ Low
Shape-constrained Bernstein polynomials §2.3.1	⚠ Limited degree	⚠ Explicit constraints, absolutely monotonic difficult	✅ Yes	⚠ Sensitive	🔵 Robust	🔵 Flexible, but affected	✅ Any	❌ Nonparametric	⚠ Affected by degree and constraints	⚠ Moderate
Precisely sampled Bernstein polynomials §2.3.1	🔵 Selected degree	🔵 Inherited from source function	✅ Yes	✅ Irrelevant	❌ Highly sensitive	❌ Fixed target	❌ Fixed	❌ Nonparametric	🔵 Moderate	🔵 Medium
Constrained Vandermonde §2.3.2	⚠ Limited degree	✅ Absolutely monotonic, by construction	⚠ Local optima	⚠ Sensitive	⚠ Sensitive	🔵 Flexible, but affected	✅ Any	🔵 Semi-parametric	⚠ Suboptimal	✅ Low
Convex splines§2.3.3	⚠ Very limited degree	⚠ Convex	✅ Yes	⚠ Sensitive	🔵 Robust	🔵 Flexible, but affected	✅ Any	❌ Nonparametric	⚠ Affected by degree and constraints	🔵 Medium
Greedy low-balling monomial sums §2.3.4	🔵 Highly flexible, constrained by number of parameters	✅ Absolutely monotonic, by construction	⚠ No	⚠ Sensitive	🔵 Moderately robust	✅ Specialised fit	✅ Any	🔵 Semi-parametric	⚠ Inefficient	✅ Low

GLAMPF is the only method that has good representation of the target function family and absolute monotonicity guarantees. The only other method with good qualities in these columns are precisely sampled Bernstein polynomials , which are completely infeasible due to fixed sampling, nonparametric nature, and no space for an asymmetric loss function. In exchange, our method does not guarantee optimality, is sensitive to hyperparameters, and is computationally inefficient. These are all issues that we are prepared to accept for our target problem.

Non-linear least squares exponential sums and Vandermonde interpolation with positivity constraints are the only two other methods with acceptable combinations of properties for our problem, and both have significant disadvantages with respect to GLAMPF.

3 Evaluation

All of the code discussed in this section can be found at `https://github.com/Undeceiver/AbsolutelyMonotonicFitting/releases/tag/20260322`

We implemented and evaluated the following of the discussed methods:

- Constrained piecewise linear interpolation §2.2.
- Constrained Vandermonde fit §2.3.2.
- Convex splines §2.3.3.
- Greedy lowballing absolutely monotonic polynomial fit §2.3.4.

In order to compare their performance, we evaluate these on multiple source functions. For each source function, we produce a random uniform sample in $[0, 1]$, add positive random noise with an exponential distribution, and then run the fitting method on the resulting sample. We do multiple runs with different random samples for each source function, and measure mean square error (normalized to the value of the function at 1, NMSE) and total soft floor loss function, as well as visualizing the result. We measure these with respect to the noisy random sample used for the fit, and with respect to an equispaced noiseless sample, to check how well the characteristics of the source function are captured versus the method overfits to the noise.

We use a broad range of source functions to test the limits of each method:

- **Absolutely monotonic functions** - The core target of this article.
  - **(Hyper)-exponentials** - Theoretically need infinite Taylor series but we should be able to approximate well with fewer parameters. Variety of more and less intense growth patterns.
    - $f_1(x) = \frac{e^x - 1}{e - 1}$
    - $f_2(x) = \frac{e^{e^x} - e}{e^e - e}$
    - $f_3(x) = \frac{e^{e^{e^x}} - e^e}{e^{e^e} - e^e}$
  - **Absolutely monotonic polynomials** - These should be the best fits. We try a wide variety of degrees to test limits.
    - $f_4(x) = x^2$
    - $f_5(x) = x^5 + 2x^4 + x^3 + x$
    - $f_6(x) = 1.2x^4 + 0.37x^{50} + 0.1x^{71}$

Figure 2: Evaluation metrics with respect to noisy sampled data.

Source function	**Piecewise linear §2.2**		**Constrained Vandermonde §2.3.2**		**Convex splines §2.3.3**		**GLAMPF §2.3.4**	
	NMSE	Loss	NMSE	Loss	NMSE	Loss	NMSE	Loss
$f_1(x) = \frac{e^x-1}{e-1}$	1.07e-3	3.61e-2	3.24e-3	6.26e-2	1.13e-3	3.71e-2	1.41e-3	3.76e-2
$f_2(x) = \frac{e^{e^x}-e}{e^{e^x}-e}$	1.02e-3	3.51e-2	**1.13e-2**	**1.17e-1**	1.06e-3	3.58e-2	2.12e-3	4.61e-2
$f_3(x) = \frac{e^{e^{e^x}}-e^e}{e^{e^{e^x}}-e^e}$	8.67e-7	1.02e-3	6.01e-3	**8.54e-2**	2.99e-3	**6.03e-2**	1.82e-5	4.26e-3
$f_4(x) = x^2$	6.21e-4	2.74e-2	3.2e-3	6.23e-2	6.27e-4	2.75e-2	7.45e-4	2.73e-2
$f_5(x) = x^5 + 2x^4 + x^3 + x$	2.82e-3	1.31e-1	1.1e-2	2.59e-1	3.04e-3	1.36e-1	7.09e-3	1.88e-1
$f_6(x) = 1.2x^4 + 0.37x^{50} + 0.1x^{71}$	1.31e-3	5.14e-2	4.64e-3	9.69e-2	2.35e-3	6.89e-2	2.51e-3	6.47e-2
$f_7(x) = x^{200}$	4.14e-8	2.24e-4	3.16e-3	**6.2e-2**	3.24e-3	**6.29e-2**	6.27e-6	2.5e-3
$f_8(x) = 3x^2 - x^3$	1.14e-3	5.26e-2	6.12e-3	1.22e-1	1.16e-3	5.3e-2	5.06e-3	1.01e-1
$f_9(x) = x^4 - 4x^3 + 6x^2$	3.08e-3	1.06e-1	1.1e-2	2e-1	3.09e-3	1.06e-1	1.23e-2	1.92e-1
$f_{10}(x) = \sin(\frac{\pi}{2}x - \frac{\pi}{2}) + 1$	9.5e-4	3.39e-2	**9.53e-3**	**1.07e-1**	9.66e-4	3.42e-2	2.85e-3	5.34e-2
$f_{11}(x) = 4x^2\{x \in [0, 0.5]\}, 4x - 1\{x \in [0.5, 1]\}$	3.12e-3	1.07e-1	1.31e-2	2.19e-1	3.1e-3	1.06e-1	1.59e-2	2.18e-1

    * $f_7(x) = x^{200}$

- **Convex but not absolutely monotonic functions** - We evaluate these to see how well methods constrained to absolutely monotonic solutions can perform in them, where convex methods' additional flexibility should be benefitial.
  - **Convex polynomials** - Limited non-zero derivatives.
    * $f_8(x) = 3x^2 - x^3$
    * $f_9(x) = x^4 - 4x^3 + 6x^2$
  - **Trigonometric functions** - Infinitely differentiable and infinite non-zero derivatives, but not absolutely monotonic.
    * $f_{10}(x) = \sin(\frac{\pi}{2}x - \frac{\pi}{2}) + 1$
  - **Piecewise polynomial** - Convex, growing, but second derivative is not continuous.
    * $f_{11}(x) = 4x^2\{x \in [0, 0.5]\}, 4x - 1\{x \in [0.5, 1]\}$

4 Results and discussion

Figure 2 summarizes the evaluation metrics with respect to the noisy sample data. Emphasized are values that are particularly poor compared to other results for the same sample. Perhaps more interestingly, figure 3 summarizes the metrics with respect to the source function. Figures 4 and 5 show this general comparison graphically on a logarithmic scale.

Figure 3: Evaluation metrics with respect to source function.

Source function	**Piecewise linear §2.2**		**Constrained Vandermonde §2.3.2**		**Convex splines §2.3.3**		**GLAMPF §2.3.4**	
	NMSE	Loss	NMSE	Loss	NMSE	Loss	NMSE	Loss
$f_1(x) = \frac{e^x-1}{e-1}$	1.05e-4	1.02e-2	7.16e-4	2.94e-2	1.76e-5	4.19e-3	7.16e-5	8.77e-3
$f_2(x) = \frac{e^{e^x}-e}{e^e-e}$	9.67e-6	3.12e-3	**6.21e-3**	**8.68e-2**	2.65e-6	1.63e-3	6.32e-4	2.52e-2
$f_3(x) = \frac{e^{e^{e^x}}-e^e}{e^{e^e}-e^e}$	6.17e-4	2.72e-2	**1.42e-2**	**1.32e-1**	8.82e-3	**1.04e-1**	3.82e-5	6.62e-3
$f_4(x) = x^2$	7.7e-5	8.77e-3	**1.11e-3**	3.67e-2	3.65e-5	6.04e-3	5.97e-5	8.46e-3
$f_5(x) = x^5 + 2x^4 + x^3 + x$	9.18e-5	2.14e-2	**4.34e-3**	1.62e-1	2.98e-5	1.23e-2	**2.97e-3**	1.22e-1
$f_6(x) = 1.2x^4 + 0.37x^{50} + 0.1x^{71}$	4.36e-5	8.83e-3	2.3e-3	6.83e-2	1.15e-3	4.82e-2	2.78e-4	2.16e-2
$f_7(x) = x^{200}$	4.25e-4	2.27e-2	**1e-2**	**1.11e-1**	**1.02e-2**	**1.11e-1**	9.13e-5	1.05e-2
$f_8(x) = 3x^2 - x^3$	8.03e-5	1.27e-2	2.67e-3	8.05e-2	2.06e-5	6.44e-3	2.02e-3	6.83e-2
$f_9(x) = x^4 - 4x^3 + 6x^2$	4.2e-3	1.12e-1	4e-3	1.2e-1	3.96e-5	1.09e-2	4.2e-3	1.12e-1
$f_{10}(x) = \sin(\frac{\pi}{2}x - \frac{\pi}{2}) + 1$	1.41e-5	3.76e-3	**5.6e-3**	**8.24e-2**	1.1e-5	3.32e-3	7.87e-4	2.81e-2
$f_{11}(x) = 4x^2\{x \in [0, 0.5]\}, 4x - 1\{x \in [0.5, 1]\}$	2.38e-5	8.45e-3	**5.08e-3**	**1.36e-1**	1.43e-5	6.54e-3	**6.54e-3**	**1.4e-1**

Figure 4: Normalized mean square error of various methods and source functions

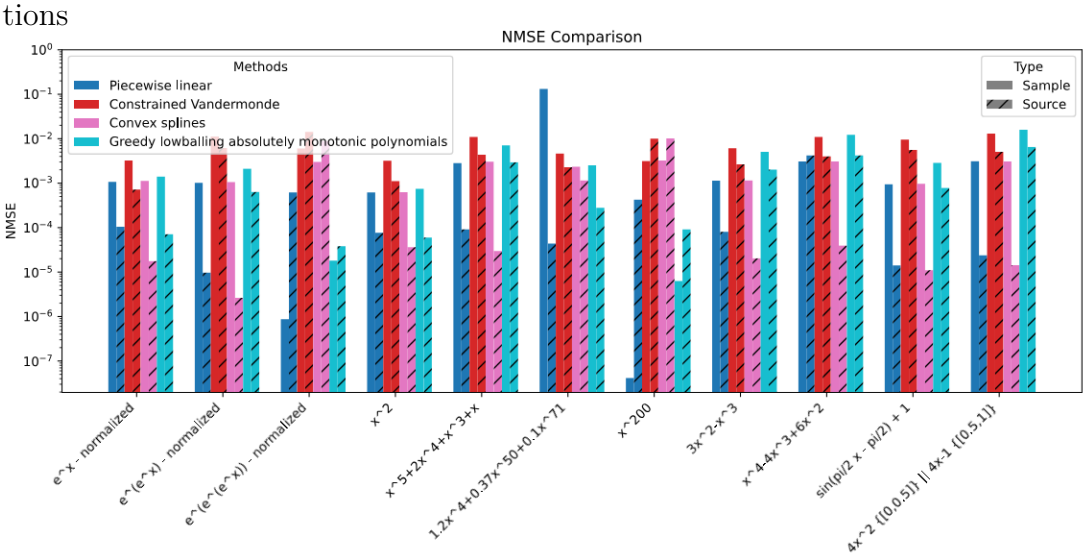
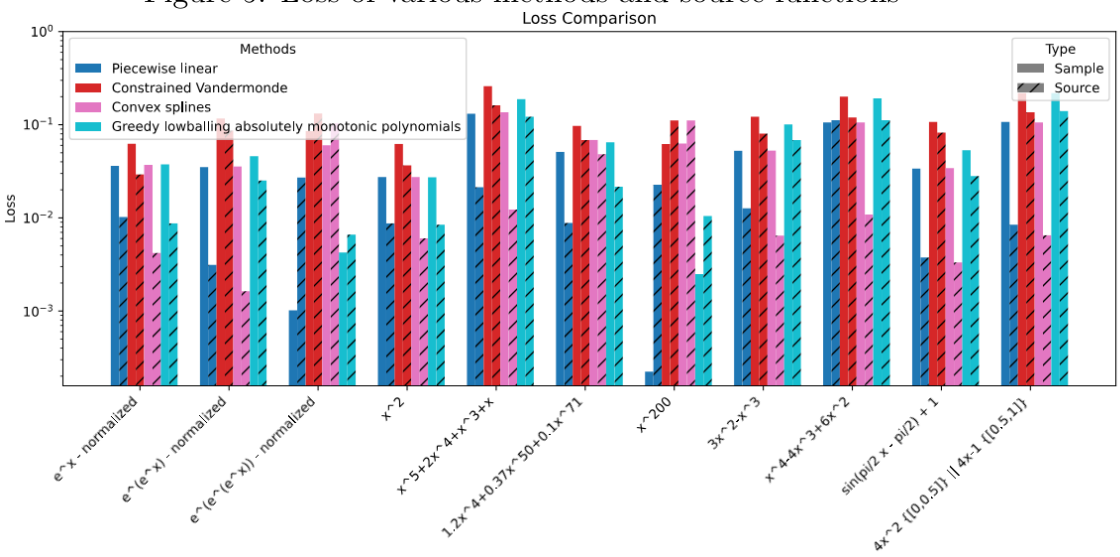


Figure 5: Loss of various methods and source functions



Piecewise linear interpolation performs quite well in some cases but is only considered as a comparison element, since it fails to satisfy two of the most core requirements of our problem statement: smoothness and absolute monotonicity. Generally speaking, the constrained Vandermonde approximation fit through non-linear least squares performs significantly worse than all other methods. Its simultaneous fitting approach struggles with the constraints and fails to fit all degree ranges simultaneously. This can be observed in test cases by noticing that it either underestimates too much on lower values, which is better approximated with low degree polynomials (see figure 6), or it underestimates too much near 1, which is better approximated with high degree polynomials (see figure 7).

With a few exceptions, evaluation on the source function is better than on the sample. This indicates that all methods are capable of capturing the underlying structure of the function and abstracting away from the noise and specific sample. The main exceptions happen with the piecewise linear approach, which is to be expected due to having a higher number of parameters and fewer constraints, and in the cases where very high degree polynomials are required to represent the underlying functions (specifically $f_3 = \frac{e^{e^{e^x}} - e^e}{e^{e^e} - e^e}$ and $f_7 = x^{200}$).

The high degree dependence is particularly noticeable in the variation across methods. Convex splines perform best of all tested methods as long as high degrees are not required, but lose significant representation capability

Figure 6: Vandermonde underestimates on lower values

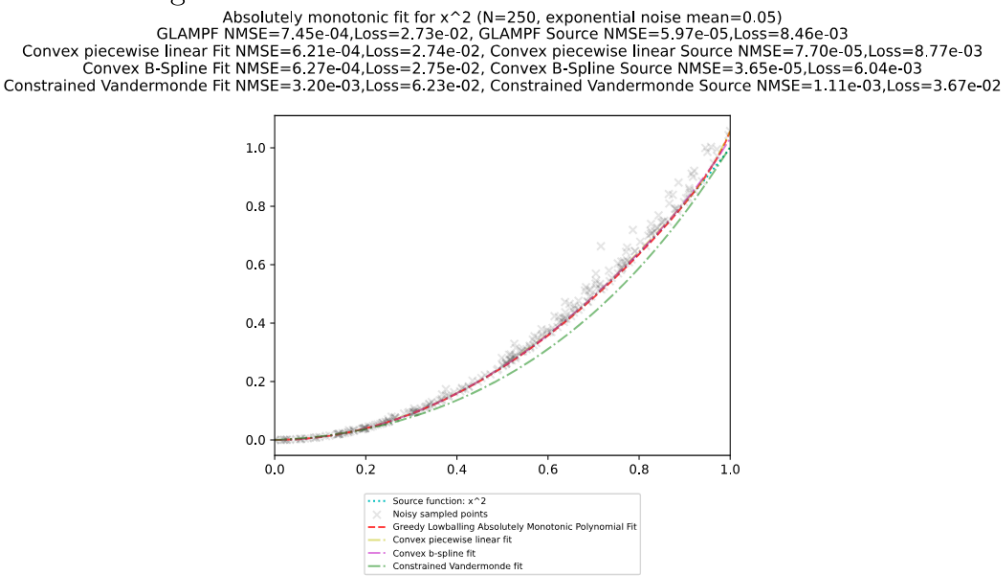


Figure 7: Vandermonde underestimates near 1

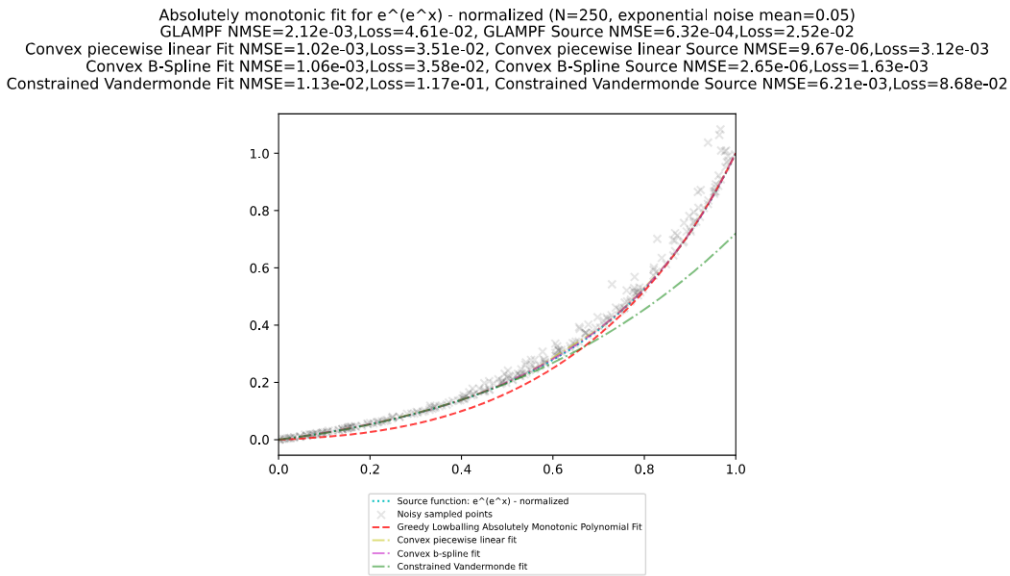
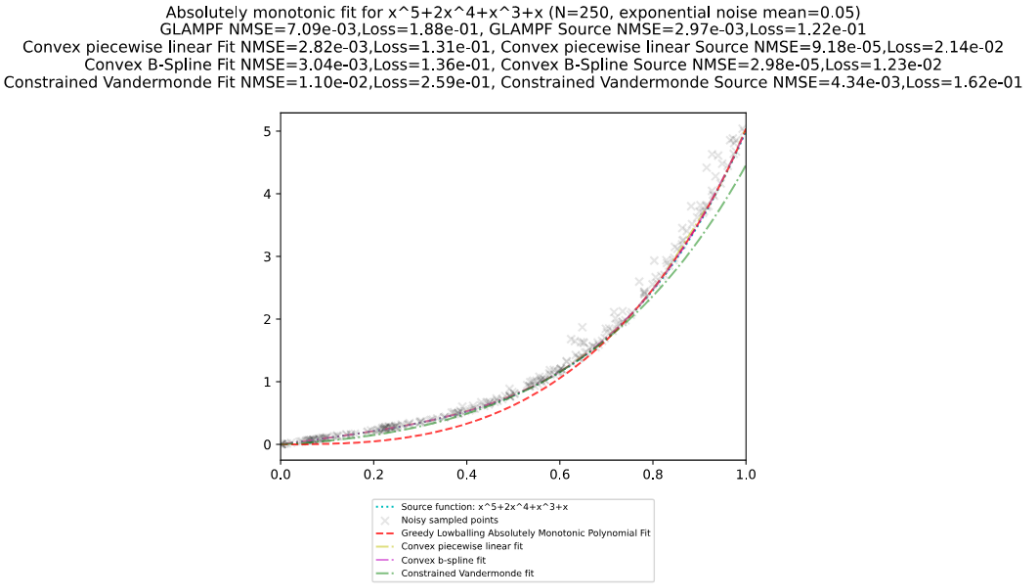


Figure 8: Convex splines perform particularly well when low degrees are sufficient



otherwise. Conversely, GLAMPF performs particularly well compared with the other methods precisely in the cases where a high degree flexibility is required, while performing reasonably well in other cases. For example, in figure 8 we can observe the dominance of convex splines over GLAMPF even in absolutely monotonic cases when low degree is sufficient. However, in very high growth cases like figures 9, 10, and 11, convex splines completely fails to fit the higher end of the curve while GLAMPF performs surprisingly well.

Unsurprisingly, GLAMPF performs significantly worse in the non-absolutely monotonic cases, consistently underestimating in lower values in order to preserve the absolutely monotonic curve shape without overestimating values near 1. See figures 12 and 13.

A consistent issue that particularly affected GLAMPF, is the overfitting to high noise values near 1 (tail). This is not an easy to solve issue, because this is fundamentally indistinguishable from high degree components in the underlying function's Taylor series. For intermediate values, noise can be avoided by bridging between lower and higher values in a consistent ways, but when there are no higher values to bridge to, the noise can easily be interpreted as an additional high degree term. This effect can be noticed for example in figure 12.

Figure 9: In very high growth cases, GLAMPF significantly outperforms convex splines

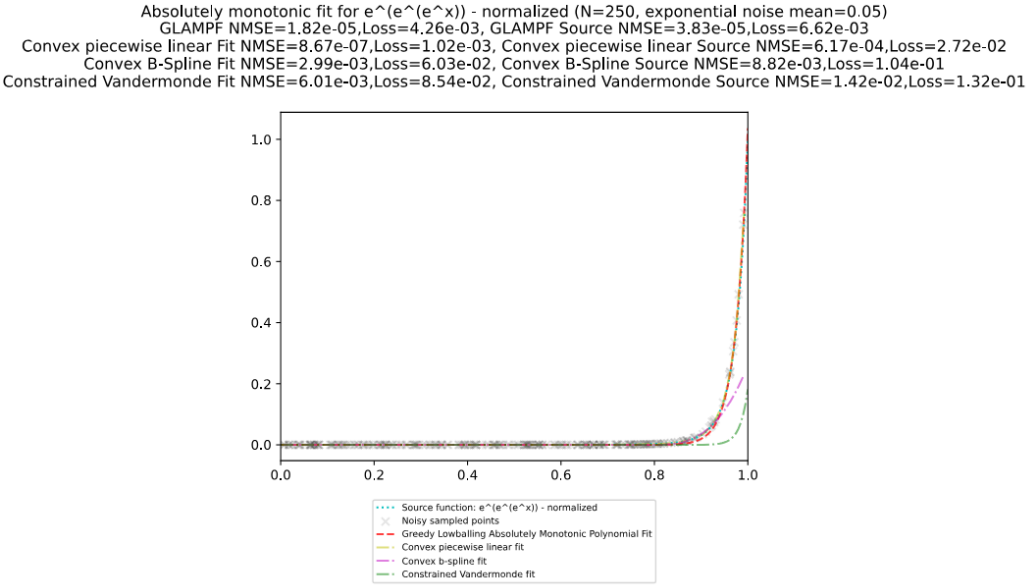


Figure 10: In very high growth cases, GLAMPF significantly outperforms convex splines

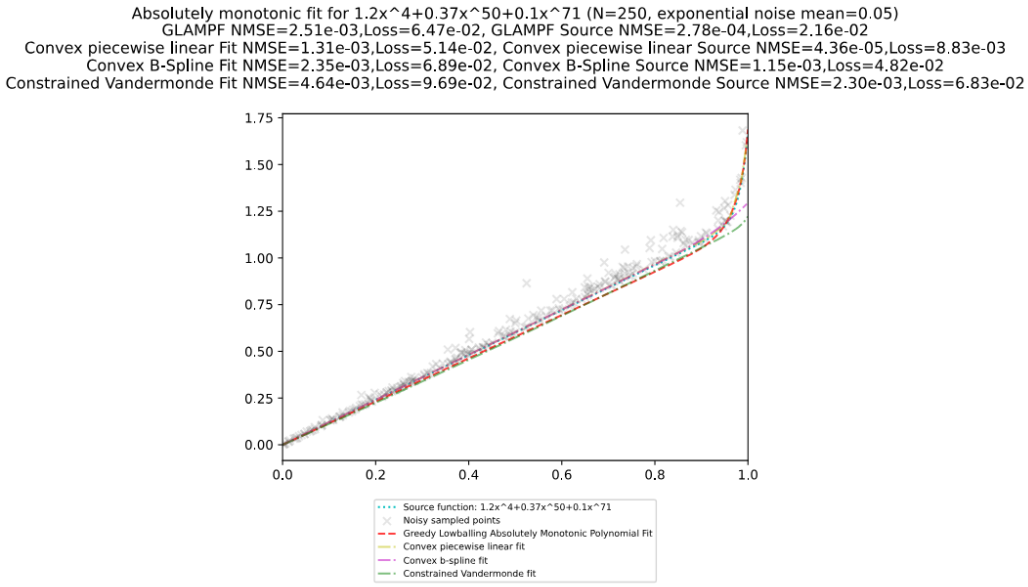


Figure 11: In very high growth cases, GLAMPF significantly outperforms convex splines

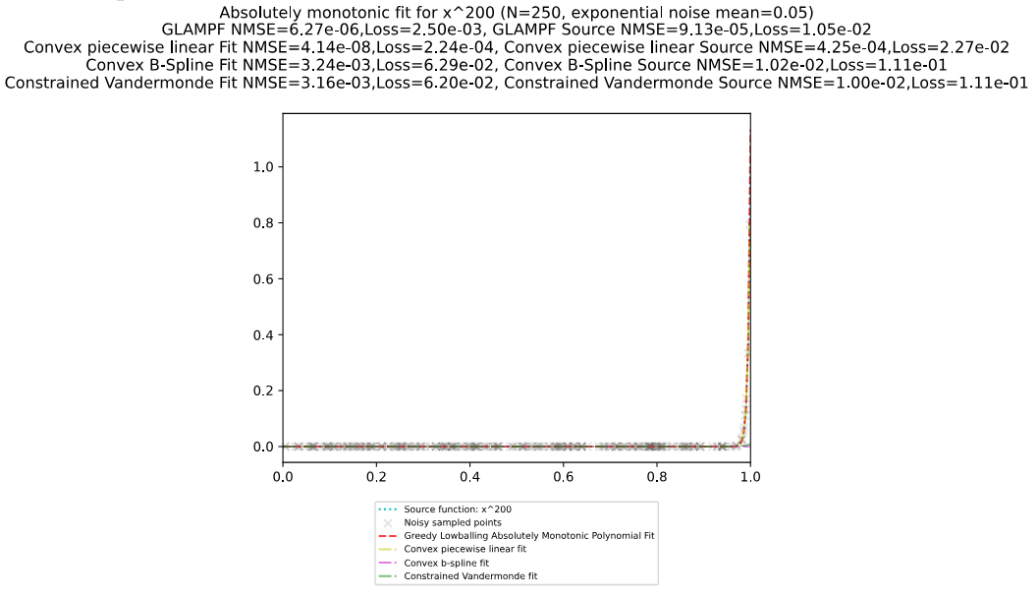


Figure 12: GLAMPF underestimates in lower values when the source function is not absolutely monotonic

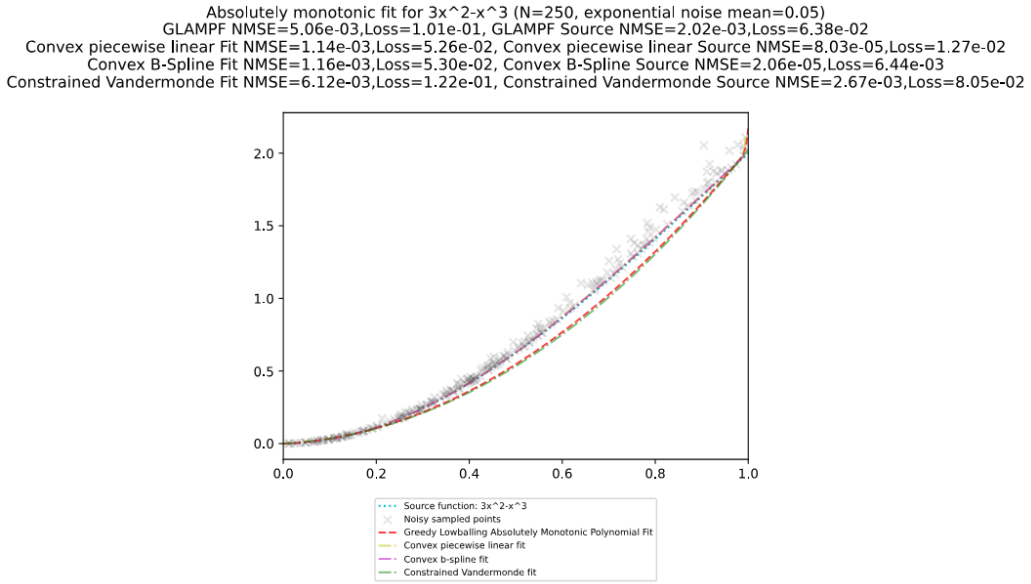
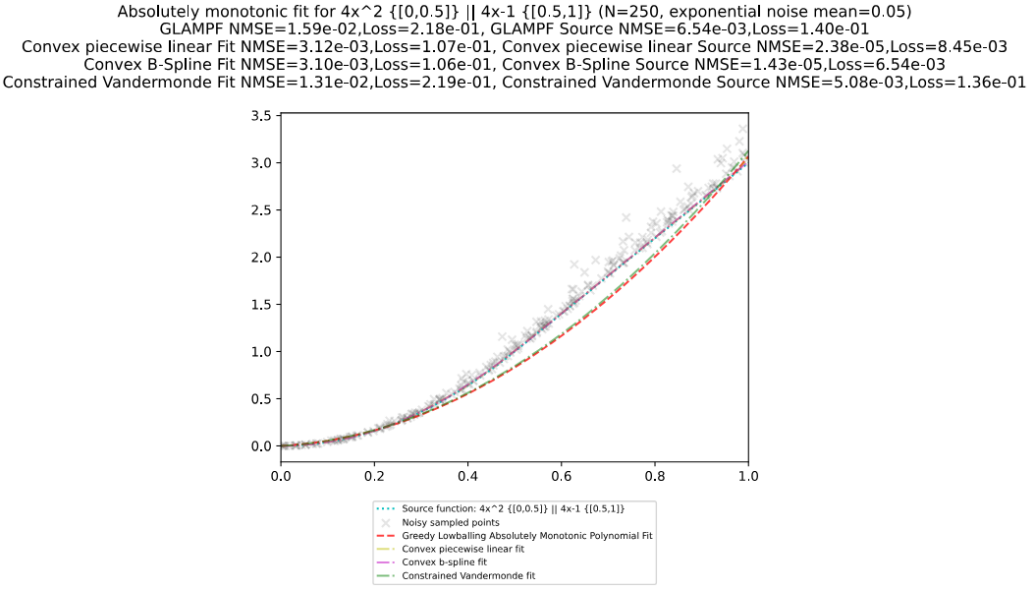


Figure 13: GLAMPF underestimates in lower values when the source function is not absolutely monotonic



Considering that, out of the tested methods, constrained Vandermonde and GLAMPF are the only two that guarantee smoothness and absolute monotonicity, we consider these results provide good evidence of the appropriateness of GLAMPF, even when in some particular cases other methods may perform better, and that GLAMPF never performs catastrophically, if the underlying source function is indeed absolutely monotonic.

5 Conclusion

In §2, we discussed the reasons why polynomial methods are preferred to other methods for fitting absolutely monotonic functions in bounded intervals. Moreover, we justified why most of the existing approaches to this problem are not very suitable. We evaluated a few of the approaches that were potentially promising. Convex splines perform extremely well as long as the underlying function does not have fast growth patterns that require high degree polynomials to represent. Our newly introduced method of greedy lowballing absolutely monotonic polynomial fitting shows significant flexibility to all types of absolutely monotonic functions and even acceptable results on other convex but not absolutely monotonic functions. It is particularly effective at detecting high growth patterns near 1. However, it does not

produce optimal results in most cases and frequently shows some level of underestimation in low or mid range values.

The significant differences in results of the various methods linked to the growth patterns suggests ideas for future work that combine the strengths of multiple approaches. For example, a hybrid algorithm that attempts multiple methods, including some not implemented here, such as exponential sums, and selects the one that performed best for the particular data presented, could potentially perform extremely well. Similarly, the greedy approach could be extended beyond polynomials to consider exponential or hyperexponential terms. Dynamically adjusting the degree of splines or Bernstein polynomials, or dynamically adjusting growth constraints in some of these methods, could lead to a broader and more effective search process that finds a more suitable representation of the data in a wider variety of cases.

The issues with the approaches described here can be summarized in two families: unsuitable constraints and assumptions for the specific underlying function, and non-optimal methods that do not produce the best results possible. Most of the approaches discussed can be clearly associated with at least one of these limitations, so the combination of both or the dynamic adjustment of constraints and assumptions depending on results could potentially find a better balance in more varied cases.

For the time being and the target problem discussed in §1.3, we believe that while sometimes suboptimal, GLAMPF offers the most robust results and strongest properties of the resulting function.

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